

Motor Imagery Classification from EEG for Brain–Computer Interfaces: A Multiscale Inception Approach

Bianciardi Tommaso [†], Zanovello Marco [†]

Abstract—Motor imagery (MI) classification from electroencephalographic (EEG) signals is a major challenge in the design of non-invasive Brain–Computer Interfaces (BCIs), where the goal is to decode neural intention and translate it into executable commands. Obtaining consistent generalization within subjects remains an open problem, as subject variability in neural rhythms hinders the direct application of fixed-kernel architectures to unseen users.

In this paper we propose a *Multiscale Inception CNN*, an architecture that employs parallel convolutional branches with different temporal kernel sizes to capture μ and β oscillations at multiple resolutions simultaneously. The model is evaluated on the public Kaya et al. (2018) dataset across a three-class MI paradigm (left hand, right hand, passive rest) under a strict cross-subject evaluation protocol.

Experimental findings demonstrate that our Inception model reaches comparable performance to the EEGNet baseline across five random seeds, showing no considerable difference between the two architectures. These outcomes suggest that our lightweight Inception module represents a viable alternative that matches EEGNet at comparable cost for real-time, subject-independent MI-BCI decoding.

Index Terms—Brain–Computer Interface, Motor Imagery, EEG, Convolutional Neural Networks, Inception Module, EEGNet

I. INTRODUCTION

Brain-computer interfaces (BCIs) based on electroencephalography (EEG) enable the user to control external devices without requiring any muscular activity. Typically, BCIs are used in medical applications such as the control of prosthetic limbs. Motor Imagery (MI) - the simulation of movement without physical execution - is a primary paradigm for neural technology, as it relies on rhythms that can be evoked voluntarily and is very useful for neurorehabilitation and human-machine interaction [1].

EEG signals exhibit a low signal-to-noise ratio due to biological interference such as muscle activity and eye blinks, as well as environmental interference such as cable artifacts. They are also non-stationary and highly variable across subjects, making it difficult for a classifier trained on one subject to generalize to another. Recent research has introduced convolutional architectures such as EEGNet [2] to address this: EEGNet uses depthwise and separable convolutions to build a single, compact Convolutional Neural Network (CNN)

that generalizes BCI paradigms while capturing well-known EEG feature-extraction concepts like spatial filtering and filter-bank construction, and it has since become a fundamental tool for EEG deep learning. Beyond EEGNet, other works have used multi-branch convolutional designs [3] such as Inception, which was originally applied to computer vision and speech recognition tasks that process input through parallel convolutions [4].

In this report, we implement both an EEGNet-style baseline and a creative Multiscale Inception CNN, evaluated under a strict cross-subject protocol on the public Kaya et al. (2018) dataset [5], assessing not only accuracy and Cohen’s Kappa, but also computational cost in terms of parameters, inference time and memory.

The remainder of this report is organized as follows. Section II reviews related work on EEG-based MI classification. Section III describes the dataset and preprocessing pipeline we applied. Section IV presents the two architectures. Section V reports and compares experimental results. Section VI concludes with limitations and future directions.

II. RELATED WORK

A. EEGNet: a Compact Baseline for EEG Classification

Lawhern et al. [2] propose EEGNet, a compact CNN built on depthwise and separable convolutions that learns frequency-specific spatial filters and temporal summaries. It shows that a single, lightweight architecture generalizes across four different BCI paradigms (P300, ERN, MRCP, SMR) and is competitive even under limited training data and without data augmentation procedures. The architecture uses Exponential Linear Unit (ELU) as activation function and sets the dropout probability to 0.5 for within-subject classification, in order to reduce overfitting, and 0.25 for cross-subject classification.

B. Inception: Multi-Scale Feature Extraction

Szegedy et al. [4] introduce the Inception architecture for image classification, whose main contribution is processing the input at parallel convolutional blocks and concatenating the results.

This approach outperforms architectures on ImageNet while using fewer parameters. The original model also relies on 1×1 dimension-reduction bottlenecks before each large convolution, needed to keep computational cost manageable when the input has hundreds of channels, as is the case

[†]Department of Information Engineering, University of Padova

[†]email: tommaso.bianciardi@studenti.unipd.it

[†]email: marco.zanovello.4@studenti.unipd.it

for natural images. EEG signals, however, have only a few dozen channels, so this kind of bottleneck is not needed. Our proposed architecture therefore keeps only the core idea of Inception – processing the input through multiple parallel convolutions at different kernel sizes – and applies it to EEG signals: three parallel temporal convolutions at different kernel sizes (16, 32, 50) capture oscillatory structure at different timescales, without the dimension-reduction bottlenecks that are not necessary at this smaller scale.

C. Inception Modules Applied to EEG

Zhang et al. [3] propose EEG-Inception, a deep CNN built on an ensemble of multiple Inception and residual modules for motor imagery classification. Evaluated on the BCI Competition IV datasets, it achieves state-of-the-art accuracy as it maintains low inter-subject variability.

Notably, our Multiscale Inception CNN converges independently toward a similar design principle, but adopts a significantly lighter architecture: a single Inception module integrated within an EEGNet-style spatial filtering framework, prioritizing parameter effectiveness and cross-subject generalization over raw performance.

III. PROCESSING PIPELINE

We use the large EEG motor imagery dataset from Kaya et al. [5]. It contains recordings from 13 different subjects, performing multiple movement tasks (CLA: left/right hand and passive; HaLT: left/right hand, left/right leg, tongue and passive) recorded from 19 scalp electrodes following the 10–20 system. Each recording is stored as a multichannel continuous time-series, in which each channel corresponds to a single electrode. These channels are combined with an additional marker channel that indicates, at each time sample, which movement the subject is performing. This extra marker channel is used later to segment the continuous signal into labeled epochs, as described in Section III-B.

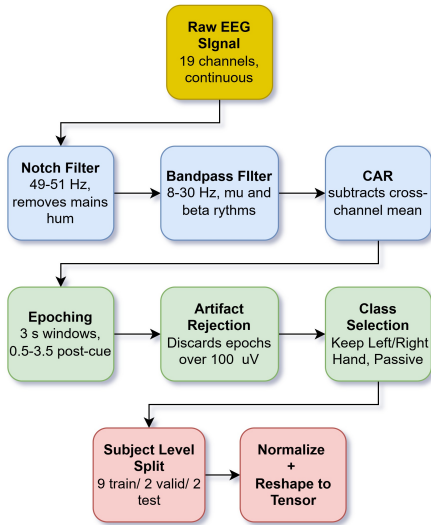


Fig. 1: Preprocessing Pipeline

The raw EEG signal is preprocessed by sequential filter, as shown in the pre-processing pipeline scheme, illustrated in Figure 1.

A. Filtering

First, we apply a Butterworth notch filter (49–51 Hz) that removes power line interference. A bandpass filter (8–30 Hz) then isolates the μ and β oscillations relevant to motor imagery. Common Average Reference (CAR) is applied by subtracting the instantaneous mean across all channels from each channel, reducing common-mode noise.

B. Epoching and artifact rejection

Then epochs are extracted using a window of $[+0.5, +3.5]$ yielding a 3-second window of neural activity. The 0.5s offset avoids contamination from the initial visual response. At a sampling rate of 200Hz, each epoch contains 600 time samples per channel, resulting in a tensor of shape (19×600) . Epochs exceeding a peak-to-peak amplitude threshold of $100 \mu\text{V}$ on any channel are discarded. This threshold effectively eliminates epochs contaminated by large muscle artifacts or movement noise, which occur frequently in EEG recordings.

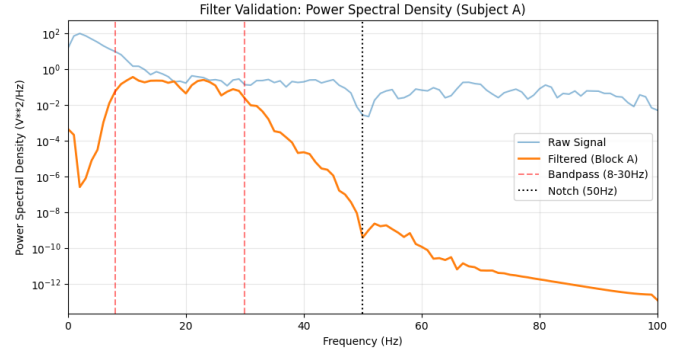


Fig. 2: Filtered signal

C. Class selection

An inspection of the dataset revealed that not all subjects followed the same experimental paradigm. As shown in Table 1, 5 of the 13 subjects contributed recordings from both CLA and HaLT, 7 subjects performed HaLT only, and 1 subject (Subject D) contributed CLA-only recordings, meaning this subject did not perform leg or tongue imagery. We note that Subject J’s CLA recording uses the Inter (Interactive Interface) acquisition mode, which differs from the rest of the dataset in signal resolution and dynamic range; the CLA session of Subject J was therefore excluded from our study to preserve signal consistency across subjects.

Because a 6-class cross-subject setup would train and evaluate on a structurally uneven distribution of paradigms across subjects, we restricted the classification problem to the three classes performed by every subject in the dataset: Left Hand, Right Hand, and Passive. CLA and HaLT recordings were then

pooled together for these shared classes. The resulting subject-level train/validation/test split (Section III-D) is randomized across this combined-paradigm subject pool.

TABLE 1: Paradigm composition and class availability per subject.

Subject	Paradigm(s)	Classes present
A	CLA + HaLT	[1, 2, 3]
B	CLA + HaLT	[1, 2, 3]
C	CLA + HaLT	[1, 2, 3]
D	CLA only	[1, 2, 3]
E	CLA + HaLT	[1, 2, 3]
F	CLA + HaLT	[1, 2, 3]
G	HaLT only	[1, 2, 3]
H	HaLT only	[1, 2, 3]
I	HaLT only	[1, 2, 3]
J	HaLT only	[1, 2, 3]
K	HaLT only	[1, 2, 3]
L	HaLT only	[1, 2, 3]
M	HaLT only	[1, 2, 3]
Summary		
CLA-only subjects: 1		
HaLT-only subjects: 7		
Mixed (CLA + HaLT) subjects: 5		

D. Dataset splitting

For the data splitting in train, validation and test sets we decided to proceed at subject level which allows to simulate a realistic cross-subject BCI scenario. This makes the classification problem inevitably more difficult but ensures that no test subject's trials are seen during training, preventing data leakage. In particular 9 subjects for training, 2 for validation and 2 for testing resulting in almost 70% – 15% – 15% split.

This subject-level partition yields 22873, 1897, and 2393 epochs for the training, validation, and test sets respectively. Despite the heterogeneity in paradigm composition across subjects, the resulting class distribution remains close to balanced in all three splits, as shown in Table 2, which we attribute to the pooling of CLA and HaLT recordings described in Section III-C.

TABLE 2: Dataset subject splitting. Number of epochs across training, validation and test sets

Split	Subjects	Epochs	LH	RH	Passive
Training	9	22,873	7,798	7,778	7,297
Validation	2	1,897	627	639	631
Test	2	2,393	811	786	796
Total	13	27,163	9,236	9,203	8,724

E. Normalization and tensor formatting

Channel-wise z-score normalization is applied as $x' = (x - \mu)/\sigma$, where μ and σ are the per-channel mean and standard deviation computed exclusively from the training set, preventing data leakage from validation and test sets. Each epoch is then reshaped to $(19 \times 600 \times 1)$ and class labels are shifted from $\{1, 2, 3\}$ to $\{0, 1, 2\}$.

IV. MODELS

Both models have similar hyperparameters, providing a fair comparison between the two implemented architectures. In both cases the input is a tensor of shape $(19 \times 600 \times 1)$ — 19 EEG channels, 600 time samples at 200 Hz, and a single feature dimension — and output a probability distribution over the three target classes via a softmax layer. All convolutional layers omit bias terms, and L2 regularization ($\lambda = 0.001$) is applied throughout. Both models are trained with the Adam optimizer minimizing sparse categorical cross-entropy loss.

A. EEGNet Baseline

Our baseline follows the EEGNet architecture proposed by Lawhern et al. [2], adapted for our sampling rate of 200 Hz. The architecture is illustrated in Figure 3.

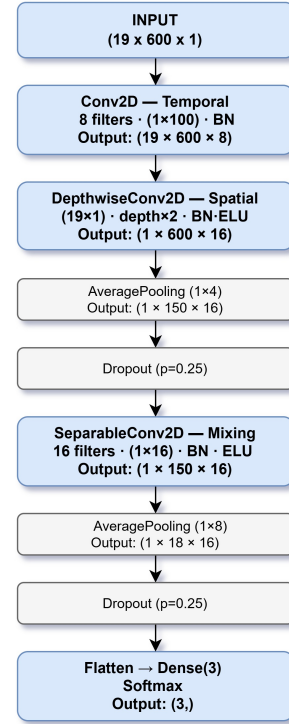


Fig. 3: EEGNet Architecture

Block 1 consists of a temporal Conv2D with 8 filters and kernel size (1×100) , where the kernel length corresponds to half the sampling rate, allowing the network to capture frequencies down to 2 Hz. Batch normalization follows. A DepthwiseConv2D with kernel (19×1) and depth multiplier 2 then learns a spatial filter across all electrodes, constrained

by a maximum norm of 1 to prevent weight blow-up. After batch normalization and ELU activation, average pooling (1×4) compresses the time dimension, followed by dropout ($p = 0.25$, as prescribed for cross-subject classification [2]).

Block 2 applies a SeparableConv2D with 16 filters and kernel (1×16), which first summarizes each feature map individually and then mixes them across channels. After batch normalization, ELU activation, and average pooling (1×8), a second dropout layer is applied. The output is flattened and passed directly to a Dense layer with 3 units and softmax activation.

B. Multiscale Inception CNN

Our proposed architecture replaces the single temporal convolution of EEGNet Block 1 with a parallel Inception module, as illustrated in Figure 4.

The motivation is that a fixed kernel size commits the network to a single temporal scale, whereas motor imagery signals span a range of oscillatory frequencies within the $8 - 30\text{Hz}$ band.

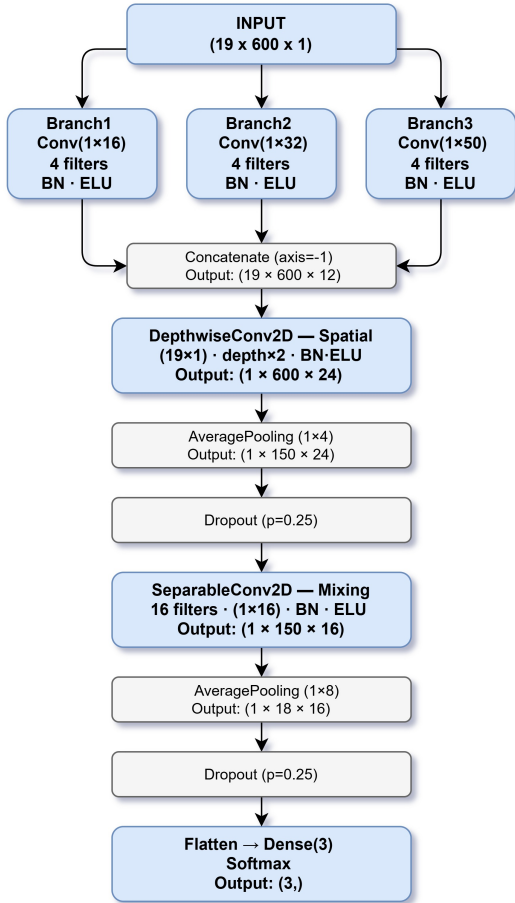


Fig. 4: Inception Architecture

The Inception module consists of three parallel Conv2D branches operating on the same input:

- Branch 1 — kernel (1×16), 4 filters: captures fine-grained temporal detail within the passband.
- Branch 2 — kernel (1×32), 4 filters: targets μ -rhythm-scale oscillations ($\approx 160\text{ms}$ window).
- Branch 3 — kernel (1×50), 4 filters: captures slower envelope modulations within the passband. Larger kernels (e.g. 128) are unnecessary given the $8\text{-}30\text{Hz}$ bandpass already applied in preprocessing.

Each branch applies batch normalization and ELU activation. The three outputs are concatenated along the filter axis, yielding 12 feature maps. The remaining architecture — depthwise spatial filter, average pooling, separable mixing layer, and classification head — is identical to EEGNet.

C. Training Protocol

We trained both models for up to 50 epochs. Early stopping monitors validation loss with a patience of 15 epochs, restoring the best weights at termination. Dropout is set to $p = 0.25$ for both models, consistent with the cross-subject setting recommended in [2]. To account for training-time stochasticity, both models are retrained across five independent random seeds; reported metrics are mean $\pm$ standard deviation over these runs.

Figure 5 shows the training and validation loss curves for both architectures under seed 42.

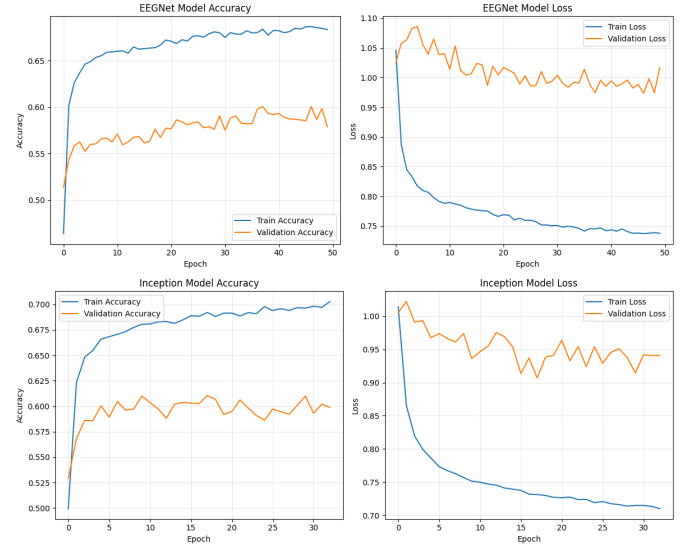


Fig. 5: Training and validation loss curves for EEGNet (top) and Multiscale Inception CNN (bottom), seed 42.

V. RESULTS

To evaluate the performance of both models, we initially examine the accuracy and Cohen’s Kappa metrics across five independent runs with different random seeds. The Kappa score is particularly relevant because it measures the agreement between model outputs and ground-truth labels, corrected for the agreement expected by chance alone. Subsequently, we investigate the class-wise performance via confusion matrices and examine the separability of the extracted features using t-SNE visualizations. Finally, we compare the computing cost of the two architectures in terms of number of parameters, model size, and inference time.

A. Classification Performance

Table 3 summarizes the mean accuracy and Cohen’s Kappa across five random seeds for both architectures. From the results of Table 3, we can see that the mean accuracies are pretty much identical, while the Inception’s standard deviation is about double that of EEGNet in both accuracy as well as Cohen’s Kappa; however, given the few number of seeds applied to both models, this difference should be interpreted with caution and would need further evaluation with other seeds in order to have a more reliable result that reflects the architecture’s property.

TABLE 3: Mean classification performance over 5 random seeds (mean $\pm$ std).

Model	Accuracy	Kappa
EEGNet baseline	0.6986 ± 0.0146	0.5482 ± 0.0216
Inception	0.6939 ± 0.0274	0.5411 ± 0.0406

Figure 6 illustrates the accuracy and Kappa scores with error bars representing the standard deviation across the seeds, confirming the stability of both training procedures.

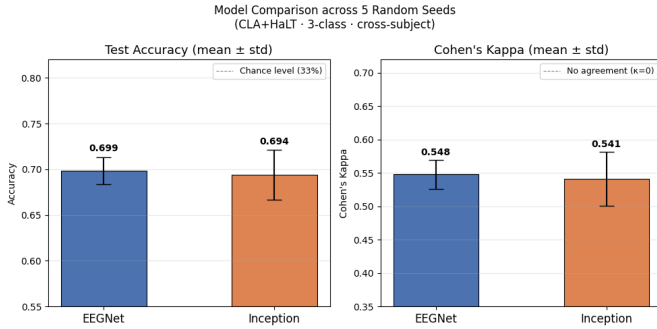


Fig. 6: Mean accuracy and Cohen’s Kappa with standard deviation across 5 random seeds for EEGNet and Multiscale Inception CNN.

B. Confusion Matrices

Figures 7 and 8 show the confusion matrices for both models under seed 42. As observed, both models offer a similar error distribution. Both architectures are reasonably robust at distinguishing active motor imagery (Left Hand vs.

Right Hand), with relatively few instances of active states being misclassified as Passive. However, both models struggle to distinguish significantly from the true Passive class, frequently misclassifying resting states as active motor intent (either Left or Right Hand). This indicates that resting-state EEG patterns are highly variable between individuals and are easily confounded with active neural rhythms in a cross-subject evaluation.

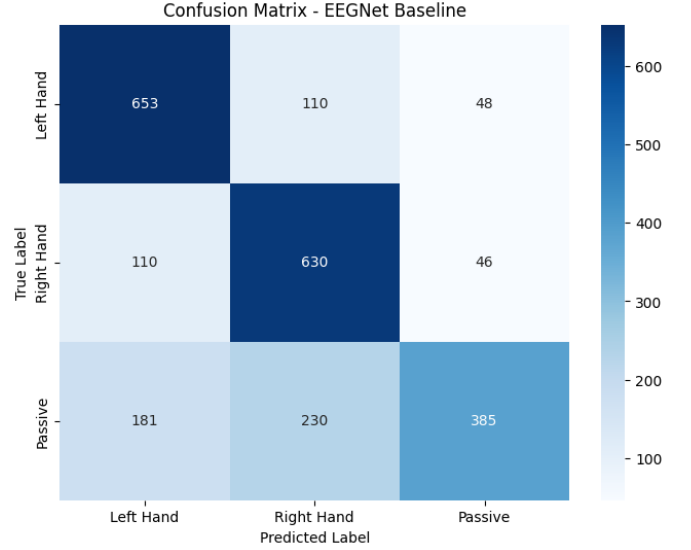


Fig. 7: Confusion matrices for EEGNet, seed 42

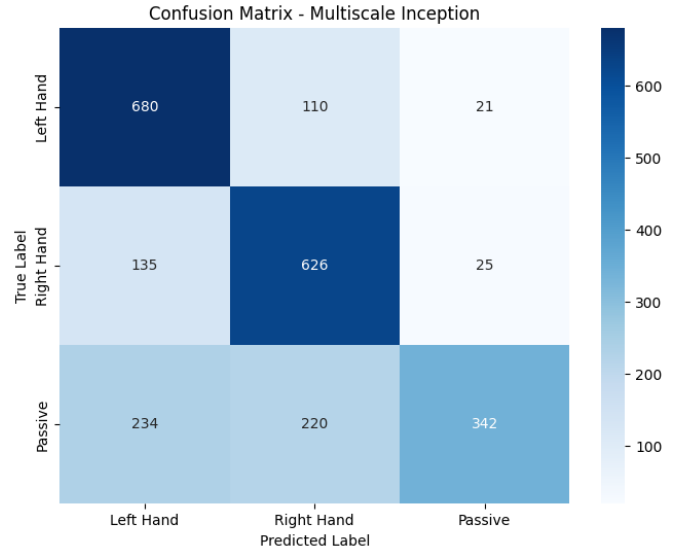


Fig. 8: Confusion matrices for Inception, seed 42

C. Latent Space Visualization

Figures 9 and 10 provide t-SNE projections of the penultimate layer activations on the test set for both models (seed 42). The visualizations indicate that both architectures learn comparable topological representations of the EEG data. In both latent spaces, there is observable clustering with the Passive state (orange) generally occupying different regional subspaces compared to the active motor classes (blue and green). However, the clusters remain highly entangled with significant overlap. This visual evidence is consistent with the $\approx 69\%$ classification accuracy, illustrating the inherent difficulty of achieving distinct linear separability among different subjects without explicit domain adaptation approaches.

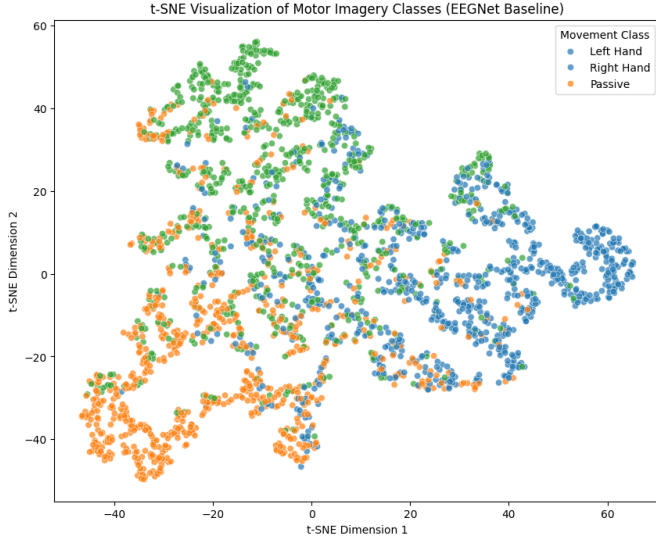


Fig. 9: t-SNE visualization for EEGNet, seed 42.

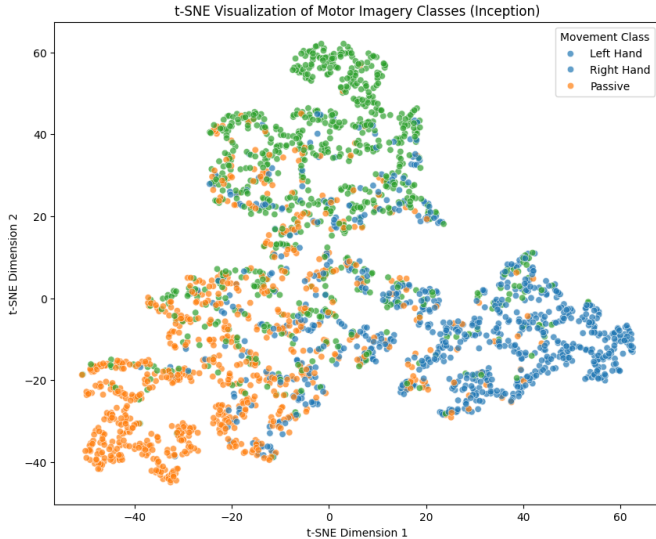


Fig. 10: t-SNE visualization Multiscale Inception, seed 42

D. Computational Complexity

Table 4 reports the number of parameters, estimated model size, and per-trial inference time for both architectures.

TABLE 4: Computing cost comparison.

Model	Parameters	Size (KB)	Inference (ms)
EEGNet	2643	10.32	0.2640
Inception	2691	10.51	0.2652

We conclude that both models share an almost identical, parameter-efficient footprint (≈ 2600 parameters and ≈ 0.26 ms inference time). Their representational capacity is practically the same: both appear capable of capturing the relevant μ and β rhythms. This efficiency is particularly notable when compared to the EEG-Inception [3], which reports around 200k parameters for binary classification and over 8 million parameters for four-class classification - about two to three orders of magnitude larger than our models. This reflects the design choice we actually applied: instead of stacking multiple Inception and residual modules as in [3], we integrate a single, lightweight Inception block within an EEGNet-style framework, prioritizing parameter efficiency and cross-subject generalization over the larger capacity of a deeper, multi-module design.

VI. CONCLUDING REMARKS

In this report, we deliver a comparative study of our Multiscale Inception CNN against the EEGNet baseline [2] for cross-subject motor imagery classification from EEG signals. Both models were evaluated on the Kaya et al. dataset [5] across five random seeds.

As shown in Section V, both architectures obtain comparable performance (accuracy $\approx 69\%$, $\kappa \approx 0.54$), well above the chance level (33.3%). These results suggest that the proposed architecture does not substantially degrade classification reliability, demonstrating that our lightweight Inception module represents a viable alternative that matches EEGNet at comparable cost for real-time, subject-independent MI-BCI decoding.

A. Limitations

The main limitation of this work is the small number of subjects (13), which constrains the cross-subject split to only two test subjects per run and makes performance estimates highly sensitive to which subjects are held out.

Furthermore, cross-subject EEG classification is still an inherently difficult problem due to the strong inter-subject variability in neural oscillation patterns; the results reported here should be interpreted within this challenging context.

B. Future Work

Several directions could extend this research. First, exploring data augmentation techniques could artificially increase the training set size, likely improving generalization. We decided not to apply data augmentation because EEG signals are already highly noisy, and because we already had more than

27000 epochs, applying artificial noise could have corrupted the true motor imagery features. As shown in Table 2, the Passive class is slightly under-balanced compared to the other two, but not that much to apply data augmentation just for that class, that's also why we decided not to implement it. Additionally, a within-subject evaluation would provide a more accurate estimation of the maximum performance achievable by these architectures, by avoiding the inter-subject variability problem. Finally, extending the classification task to the full 6-class HaLT paradigm, ideally with a larger and more homogeneous dataset, would allow for a more detailed assessment of the Multiscale Inception design.

REFERENCES

- [1] J. R. Wolpaw, N. Birbaumer, D. J. McFarland, G. Pfurtscheller, and T. M. Vaughan, "Brain-computer interfaces for communication and control," *Clinical Neurophysiology*, vol. 113, no. 6, pp. from page 767 – 791, Jun. 2002.
- [2] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, "EEGNet: a compact convolutional neural network for EEG-based brain-computer interfaces," *Journal of Neural Engineering*, vol. 15, no. 5, p. 056013, Jul. 2018.
- [3] C. Zhang, Y.-K. Kim, and A. Eskandarian, "EEG-inception: an accurate and robust end-to-end neural network for EEG-based motor imagery classification," *Journal of Neural Engineering*, vol. 18, no. 4, p. 046014, Mar. 2021.
- [4] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and A. Rabinovich, "Going deeper with convolutions," in *2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2015, pp. 1–9.
- [5] M. Kaya, M. K. Binli, E. Ozbay, H. Yanar, and Y. Mishchenko, "A large electroencephalographic motor imagery dataset for electroencephalographic brain computer interfaces," *Scientific Data*, vol. 5, no. 1, p. 180211, Oct. 2018.